Learning What to Keep by Seeing What to Leave: Counterfactual Video Summarization

SCSL-SGC: Unsupervised Video Summarization via Vectorized Counterfactual Attribution

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What is Video Summarization?

- **The Core Task**: Convert a long video $\mathcal{V}$ into a short, representative summary $\mathcal{S}$ containing the most critical events.
- **Target Length**: Typically restricted to $\leq 15\%$ of the original video duration.
- **Unsupervised Setting**: No human annotated summary templates are available during training.

Summarization Concept

Full Video (N frames)

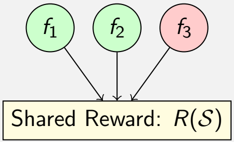
$\downarrow$ Selection Policy π_θ

Summary ($K \leq 0.15N$)

The Core Problem: High Variance Credit Assignment

- In standard RL, the model generates an entire summary $\mathcal{S}$ and receives a single global reward $R(\mathcal{S})$.
- **The Credit Assignment Problem**:
  - If a summary contains 10 good frame picks and 1 bad pick, the global reward drops.
  - All selected frames, including the 10 good ones, are penalized equally.
  - Leads to high gradient variance and slow convergence.

Uniform Credit Assignment (Bad)



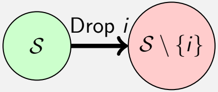
The Solution: Leave-One-Out (LOO) Attribution

- Measure the marginal contribution of each frame by removing it from the selected summary:

$$\mathcal{A}(i) = R(\mathcal{S}) - R(\mathcal{S} \setminus \{i\})$$

- **How it guides the policy**:
  - Highly informative frames receive strong positive rewards.
  - Redundant or harmful frames receive zero/negative attribution.
  - Drastically accelerates convergence.

Attribution Principle



$$\mathcal{A}(i) = R(\mathcal{S}) - R(\mathcal{S} \setminus \{i\})$$

Attribution = Marginal Gain

Proposed Model Architecture (SCSL-SGC)

Our model maps visual frames to selection probabilities in a single forward pass:

GoogLeNet Features → Linear + LN (512-dim) → 2-Layer Bi-LSTM → 2x Self-Attn (8 Heads) → MLP Score Predictor → Sigmoid Selection p_t

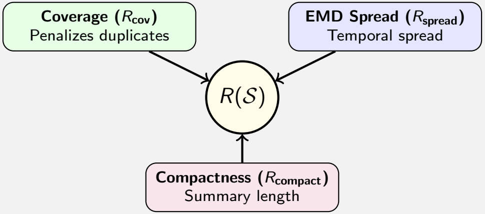
- **2-Layer Bi-LSTM**: Encodes local sequential dependencies.
- **2x Multi-Head Self-Attention Blocks**: Capture global correlations across long temporal ranges (e.g., matching start and end events of a video).

Multi-Factor Unsupervised Reward Formulation

$$R(\mathcal{S}) = 0.35R_{\text{cov}} + 0.20R_{\text{spread}} + 0.15R_{\text{compact}}$$

- We train the model purely from rewards without labels.
- Each sub-reward targets a specific physical property of an ideal summary.

Reward Components



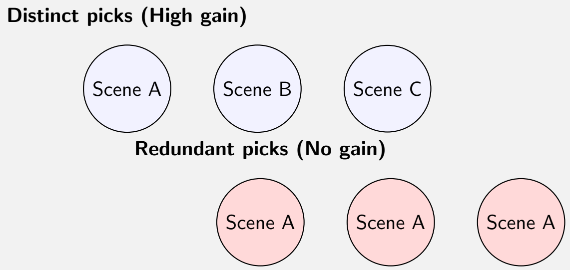
Representativeness: Submodular Coverage

- **The Core Metric**:

$$R_{\text{cov}} = \frac{1}{|V|} \sum_{i \in V} \max_{j \in \mathcal{S}} \text{sim}(i, j)$$

- Under a submodular facility-location model, the marginal gain of adding a frame drops as the summary grows.
- Once frame A is selected, selecting another frame from the same scene adds almost zero reward, forcing diversity.

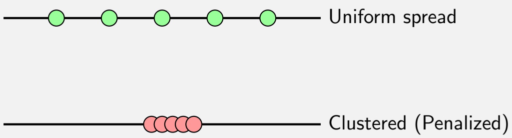
Submodularity Concept



Diversity: Earth Mover's Distance Spread

- Standard summarization models often cluster all keyframes within a single high-action event (e.g. at the middle of the video).
- **EMD Spread Reward**:
    - Measures the L_1 deviation of selected frame indices from a uniform quantile grid.
    - Penalizes clustered selections and rewards temporal distribution.

Temporal Spread vs. Clustered Selections



Vectorized $O(1)$ Optimization for Real-Time Execution

- **Computational Bottleneck**: Naively computing $R(\mathcal{S} \setminus \{i\})$ for each selected frame $i \in \mathcal{S}$ requires $O(k)$ loops, invoking the reward function repeatedly. This results in slow training ($\sim$4.5 minutes/epoch on CPU).
- **Our Fully Vectorized PyTorch Reformulation**:
  - We represent subset removals as a binary matrix mask of shape $(k \times N)$.
  - Pairwise similarities and cumulative sums are calculated in a single parallel tensor pass.
  - Resolves all attributions in $O(1)$ batch operations.

Impact on Training Throughput (CPU Execution)

Implementation	**Epoch Duration**	**Training Speedup**
Loop-based $O(k)$ Rewards	270 seconds	$1.0\times$
Vectorized $O(1)$ Rewards (Ours)	**20 seconds**	**$13.5\times$ to $15.0\times$**

Experimental Benchmarking on SumMe

Our lightweight, visual-only model compared against the latest 2024–2026 global SOTA frameworks:

Model / Configuration	SumMe F-score	Key Characteristics
Simplified GAN (arXiv:2407.04258)	51.2%	Alternating GAN training, visual-o
Semantic Gen Autoencoder (arXiv:2601.14773)	53.2%	Masked frame reconstruction, visua
Multimodal Transformer (arXiv:2505.23268)	56.4%	Visual + audio + transcripts cross-
SCSL-SGC (Ours, Seed 42)	**57.5%**	Visual-only, Counterfactual RL
SCSL-SGC (Ours, Seed 123)	**60.7%**	Visual-only, Counterfactual RL

- Outperforms the heavy multimodal transformer (arXiv:2505.23268) by **+4.3%** on SumMe.
- Achieves this using a purely visual backbone, bypassing text transcription and audio feature pipelines.

Conclusions & Takeaways

Key Contributions

- **Gradient Variance Resolution**: Showed that Counterfactual Attribution (LOO) successfully mitigates reward assignment variance in reinforcement learning for video summarization.
- **Submodular Selection**: Demonstrated that modeling coverage via a submodular facility-location reward naturally ensures representativeness and diversity.
- **Vectorized Framework**: Enabled high-throughput, CPU-friendly training of video summarizers.

Future Directions

- Extension of the vectorized counterfactual pipeline to long-form, hour-long video summarization benchmarks.
- Porting the counterfactual attribution paradigm to zero-shot selection using pre-trained Vision-Language Models (VLMs).